2-2 STEM-AND-LEAF DIAGRAM

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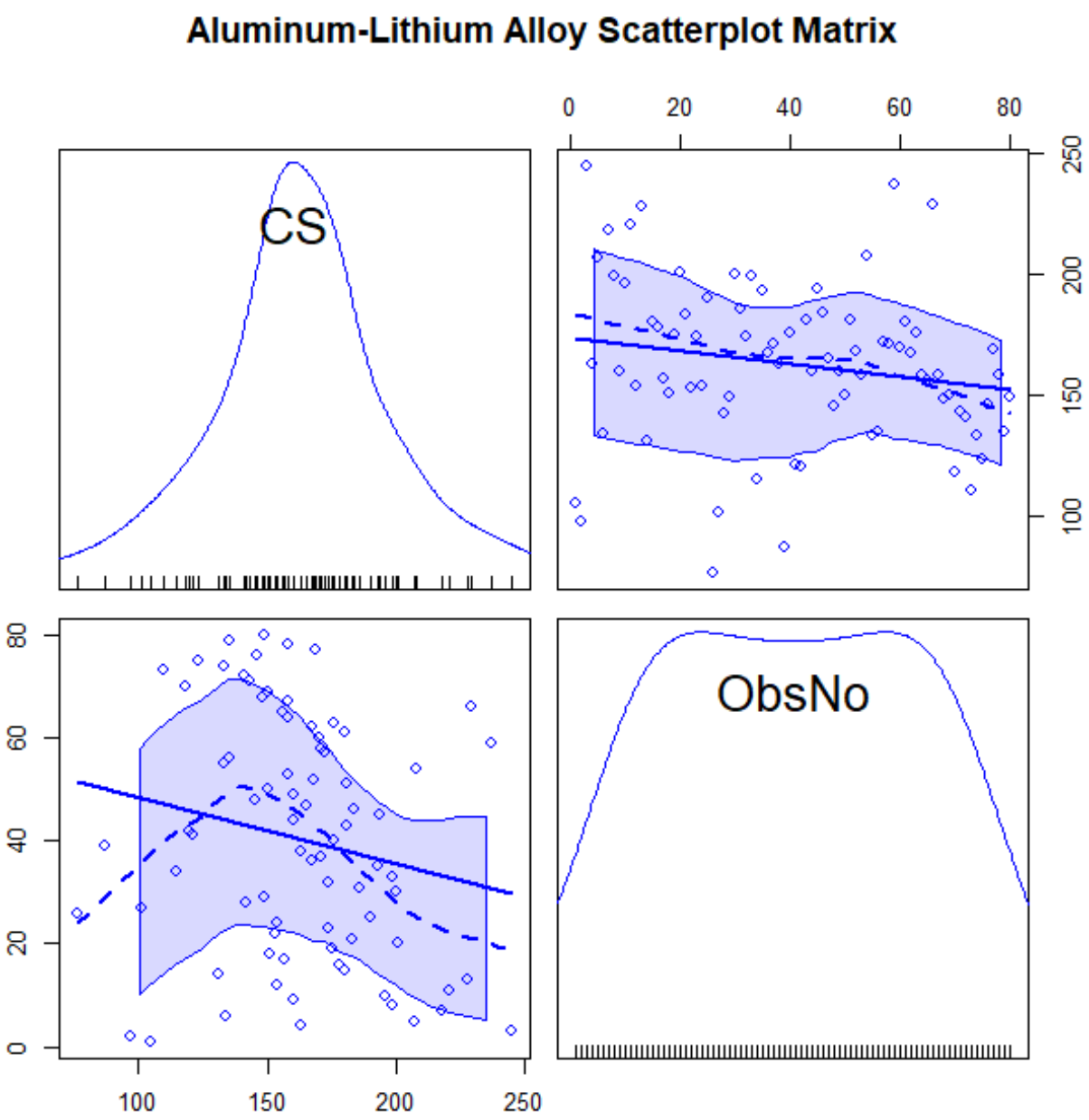
- **Dataset Overview**: **Table 2-2** presents the compressive strength (in psi) for 80 specimens of an aluminum-lithium alloy intended for aircraft structures.

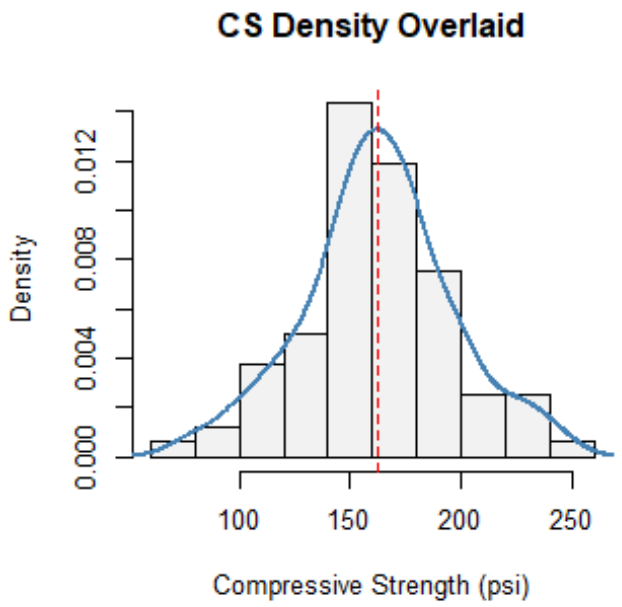
- **Format Limitations**: Because the data in **Table 2-2** is listed in the order of testing, it is difficult to extract meaningful information or answer specific questions about failure percentages.

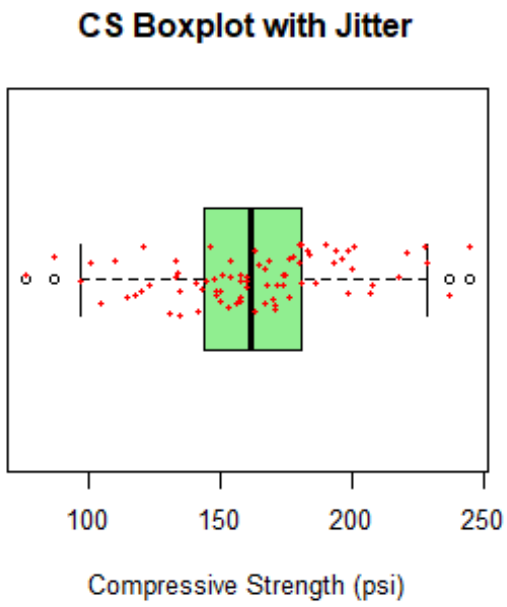
- **Visualization Challenges**: Due to the large number of observations, a dot diagram is inefficient; large datasets like those in **Table 2-2** require more effective display methods.

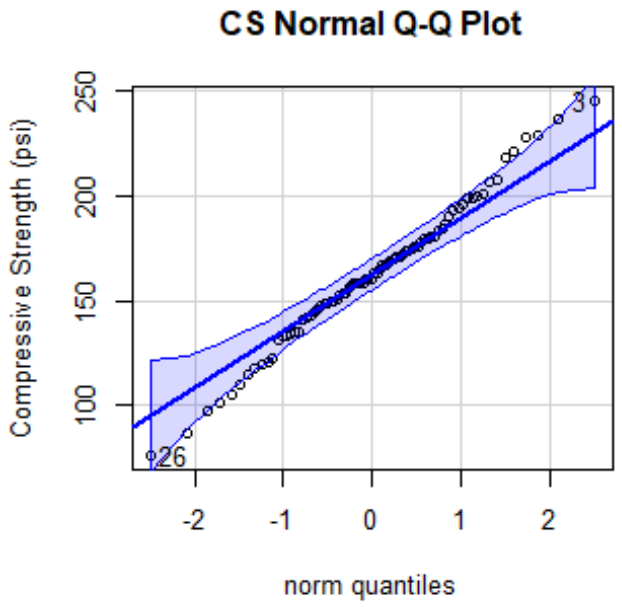
2-2 STEM-AND-LEAF DIAGRAM



2-2 STEM-AND-LEAF DIAGRAM







2-2 STEM-AND-LEAF DIAGRAM

- **Steps for Constructing a Stem-and-Leaf Diagram**: This process provides a clear visual summary of a data set where each value has at least two digits.

- **Step 1 (Partitioning)**: Separate each observation into a **stem** (the leading digits) and a **leaf** (the final digit).

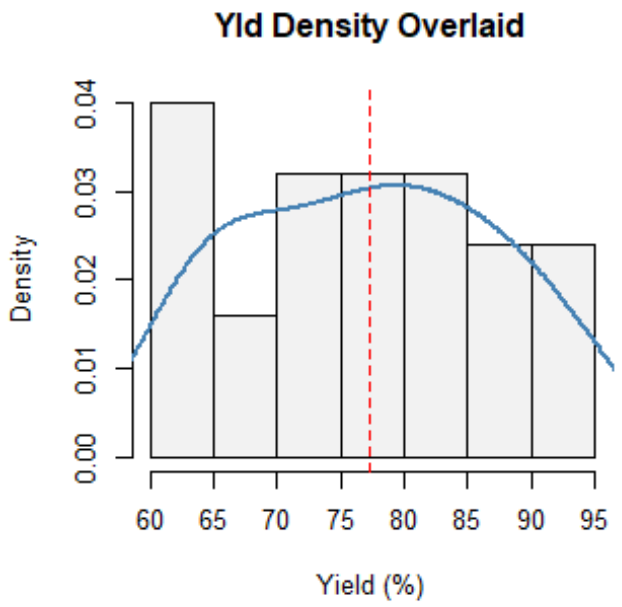
- **Step 2 (Ordering)**: Arrange the **stem** values in a vertical column.

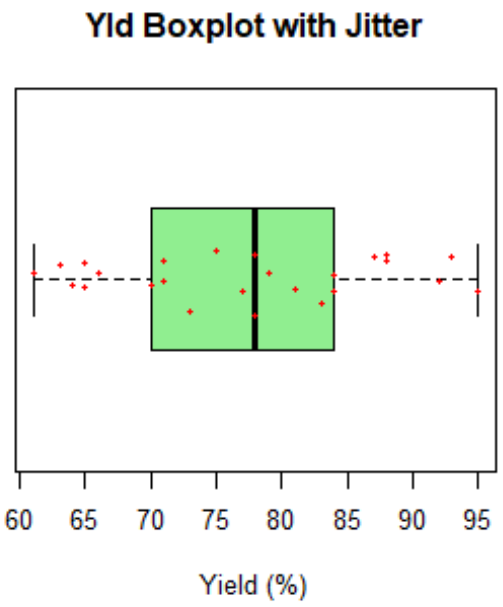
- **Step 3 (Data Entry)**: Place each **leaf** next to its corresponding **stem**.

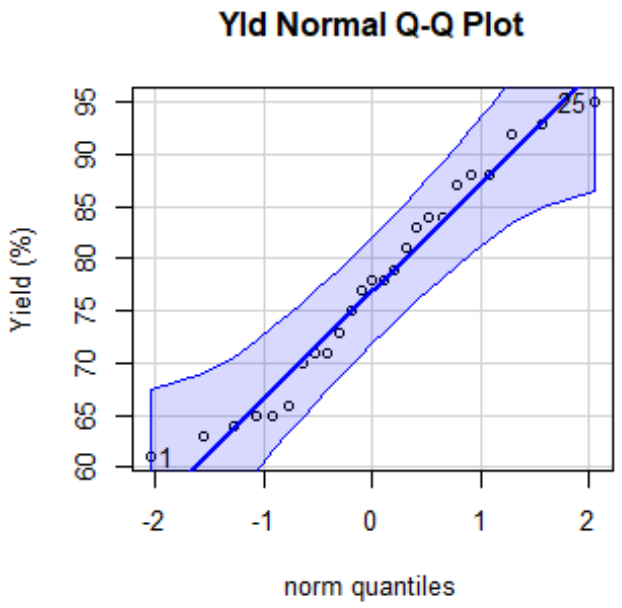
- **Step 4 (Labeling)**: Explicitly note the units used for both the **stem** and the **leaf** on the **stem-and-leaf diagram**.

- **EXAMPLE 2-4 Compressive Strength** demonstrates construction of a stem-and-leaf diagram using alloy strength data, with the resulting plot shown in **Fig. 2-4**. The diagram provides a quick analysis of the data, revealing that strengths are symmetrically distributed, with most values between 110 and 209 psi and a central value in the 150–159 psi range.

- **EXAMPLE 2-5 Batch Yield** illustrates how the selection of stems in a stem-and-leaf plot affects the visualization of 25 chemical process observations. **Figure 2-5** shows that using 6, 7, 8, and 9 as stems (2-5a) provides too little information, whereas dividing each stem into two parts (2-5b) adequately represents the data, and dividing into five parts (2-5c) results in too many stems.







2-2 STEM-AND-LEAF DIAGRAM

- **Figure 2-6** depicts a Minitab-generated stem-and-leaf display of compressive strength data, featuring leaves ordered from smallest to largest, which is termed an **ordered stem-and-leaf diagram**. Key, automated features include the use of consistent stems, an additional column providing observation counts for each stem, and the efficient ordering of leaves, a task often time-consuming to perform manually.

- An **ordered stem-and-leaf diagram** produced by Minitab, as shown in **Figure 2-6**, simplifies the identification of key data features:

  - **median**: A measure of central tendency that splits the data into two equal halves. For the 80 observations in **Figure 2-6**, it is calculated as 161.5 (the average of the two middle values).

  - **range**: A measure of variability calculated by subtracting the minimum value from the maximum value. In this example, the result is 169.

- **The interquartile range is a measure of variability**, defined as the difference between the upper and lower quartiles. **Quartiles** divide ordered data into four equal parts, with the **first** or **lower quartile** (q_1) marking the 25th **percentile**, the **second quartile** (q_2) acting as the median, and the **third** or **upper quartile** (q_3) marking the 75th **percentile**.